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Home energy efficiency evaluation based on NILM

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Abstract

This paper proposes a home appliance energy efficiency assessment method based on non-intrusive load monitoring (NILM). For the home appliance energy efficiency assessment, firstly, the NILM technology is used to analyze the user side power consumption data, and the power consumption details of the various appliances inside the user's home are obtained, from which the user's power consumption characteristics are analyzed. Then, according to the purpose and working mechanism of the electrical appliances, the household electrical appliances are classified, and different energy efficiency evaluation indexes are established according to different types of household electrical appliances. Finally, quantify the less energy-efficient appliances, determine the inefficient running time of the appliance, and calculate the electrical energy lost by the appliance in the corresponding time, and propose targeted power consumption to the user to maximize the user's power usage behavior. To realize the interaction between the user and the power grid, and greatly improve the user experience.

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1. Introduction

With the rapid development of social economy and the continuous improvement of living standards of users, as well as the frequency and duration of electricity consumption, household energy consumption accounts for an increasing proportion of the total electricity consumption. However, there is no breakthrough research on the energy efficiency assessment and analysis of household appliances in China. Therefore, it is necessary to design more

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comprehensive and effective methods for evaluating the energy efficiency of household appliances, which can provide efficient electricity consumption plan and optimize the residents' electricity consumption behavior.

Reference [1-3] elaborated the household intelligent electricity consumption strategy but is mainly based on the total amount of power consumption. Reference [4-5] mainly describes the research on energy efficiency evaluation of major power customers. However, its indicator selection is highly subjective and lack energy efficiency assessment. Reference [6] proposes a smart home management control scheme for household, but the relationship between user comfort and electrical appliances is not taken into consider. Reference [7] describes the research and application of user energy efficiency analysis system. This article is mainly aimed at enterprise users. Reference [8] puts forward an evaluation method of household appliances, but only divides household appliances into three efficiency types.

To solve the above problems, this paper proposes a home energy efficiency assessment method based on non-intrusive load monitoring (NILM). Firstly, the user electricity consumption data is analyzed using NILM technology, and analyze the characteristics of the consumption. Then, household appliances are classified and setting up different energy efficiency evaluation indicators for different types of appliances. Finally, the low energy efficiency of electrical appliances was quantified to determine the invalid running time of electrical appliances and trying to greatly improve the user's energy use experience.

2. Detailed electricity consumption data acquisition

Refined electricity consumption data includes two aspects. First, it is the data of users' electricity demand in a specific environment to meet their living and working needs. it includes household environmental data, preset values of electrical appliances (e.g., preset temperature of air conditioner at 25°C), etc. Second, it is the details of electrical appliances in the home, including real-time status, switching time and use order. It is the basis of home energy efficiency estimation.

2.1. Demand data

The main way to obtain electricity demand data is to combine the weather data of the user's home location and make statistics on his/her electricity consumption activities. Table 1 shows the statistical results of a certain user's household electricity consumption activities on working days. The temperature value represents the outdoor temperature of the family, and the light value represents the brightness of the family. As shown in the table, certain electrical activities are bound to have corresponding electrical work, and the period of electrical use is also regular. However, some appliances are related to the user's electricity consumption expectation and environmental quantity.

2.2. Electrical data

In consideration of economy and extensibility, NILM technology was used to obtain details of electricity consumption of various appliances in the user's home. NILM technology can obtain itemized data of electrical appliances only using total household electricity consumption information. It has the advantages of easy implementation, low cost, and good data integrity [9-10].

In the aggregate information of users. The active power of each electric appliance satisfies the superposition. In the process of operation, electrical appliances will leave marks on the total amount information of users. These load marks can be combined with NILM technology to realize the decomposition of total load of users and obtain the electricity data of each electrical appliance, such as active power curve, etc., as shown in Figure 1.

Table 1. Customers' activities with electrical appliances on workday

Electrical activity	Appliances	time	temperature (°C)	light
Breakfast	Lighting, kitchen appliances	06:00-07:00	25	Dark
Lunch	Kitchen appliances, air conditioning	12:00-13:00	32	Bright
Entertainment Rest	TV, air conditioning	13:00-13:30	33	Bright
Dinner	Kitchen appliances, air conditioning	17:30-18:30	31	Bright
Entertainment Rest	TV, air conditioning, lighting	20:30-21:30	30	Dark
Washing	Lighting, water heater	21:30-22:00	29	Dark
Sleep	Air conditioning	22:30-24:00	29	Dark
	none	24:00-06:00	24	Dark

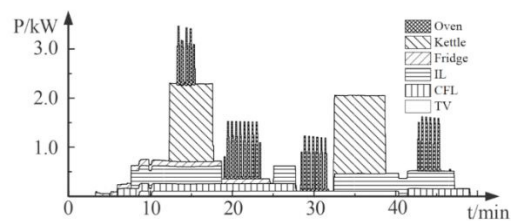


Fig. 1. NILM diagram

3. Home energy Efficiency Assessment

Based on the detailed electricity consumption data, household appliances are classified according to their working mechanism and purpose of use. And the environment parameters are also considered. Finally, the home energy efficiency evaluation is carried out in combination with the real-time running state of electrical appliances.

3.1. Classification of household appliances

This paper divides household appliances into four categories: base load, interruptible load, transferrable load, and energy storage load. Base loads are some household appliances with fixed use time, such as rice cookers and induction cookers. Interruptible load means that the status of load is greatly affected by environmental factors, such as air conditioning, electric fans, etc. Transferrable load refers to electrical appliances, such as washing machines, that can be shifted to other time periods according to the needs of users. Energy storage load refers to an electric device (such as an electric car, an electric bicycle, etc.) that first stores electric energy and then works within a specific time. The total amount of electric energy consumed during the working period can be set by the user in advance according to the next electricity consumption scenario [11].

3.2. Energy efficiency function of household appliances

The energy efficiency function of household appliances represents the quantitative relationship between the benefit of the appliance and the output value of the consumed electric energy. In order to achieve scientific and accurate energy efficiency assessment, this paper establishes different energy efficiency functions for the main load categories based on the above classification results of household appliances.

Comprehensively consider the load characteristics of four load types. In these load types, basic loads, and transferable loads, such as televisions, electric water heaters, etc., depends entirely on the user's own electricity demand. The user's electricity benefit is reflected in the use time of the electrical appliances, so the time energy efficiency function is established. However, Interruptible loads are closely related to the environment, such as

electric fans, air conditioners, etc. The user's electricity benefit is reflected in temperature and illumination, so the temperature energy efficiency function and the illumination energy efficiency function are established.

In conclusion, the energy efficiency function of household appliances is specifically divided into time energy efficiency function, temperature energy efficiency function and illumination energy efficiency function.

1) Time energy efficiency function

The time energy efficiency function is oriented to the basic load and transferrable load such as TV set and electric water heater. It can evaluate the time energy efficiency of household appliances. For any moment $t \in [t_{min}, t_{max}]$, the time energy efficiency function is shown in Formula (1), where t_s is the beginning time of the operation of the electrical appliance, and t_e is the end time of the operation of the electrical appliance.

$$E_i^t(t) = \begin{cases} 1 & t \in [t_{min}, t_{max}] \\ 0 & else \end{cases} \quad (1)$$

Where $E_i^t(t)$ represents the time energy efficiency function of electrical appliance i ; $[t_{min}, t_{max}]$ represents time an appliance has been in use; $[t_s, t_e]$ represents the running time of the appliance, which satisfies $[t_{min}, t_{max}] \subseteq [t_s, t_e]$.

For example, when the user watches TV from 19:00 to 19:30, but the actual running time is 18:50-19:30, the energy efficiency function of TV is 0 during 18:50-19:00, and the energy efficiency function of TV is 1 during 19:00-19:30.

2) Function of temperature energy efficiency

The temperature energy efficiency function is aimed at interruptible loads related to the operating state of electrical appliances and the temperature parameters of the user's home area, such as air conditioner and electric fan.

The temperature energy efficiency function is shown in Equation (2).

$$E_i^c(T_t) = \begin{cases} 0 & T_t < T_{min} \\ 1 & T_t \in [T_{min}, T_{max}] \\ 0 & T_t > T_{max} \end{cases} \quad (2)$$

Where $E_i^c(T_t)$ represents the temperature energy efficiency function of electrical appliance i ; $[T_{min}, T_{max}]$ is the threshold of expected temperature; T_t is the temperature of the user's home area at time t . If T_t is in range $[T_{min}, T_{max}]$, the temperature utility function is 1, otherwise it is 0.

Taking air conditioning as an example, considering the economy and extension of the method, the temperature parameter of the home area is not the user's indoor temperature but the real-time temperature obtained by the meteorological bureau. When the target temperature is $[20, 25]$ °C and the outdoor temperature is 23 °C, then the energy efficiency function of the air conditioning is 1.

3) Illumination energy efficiency function

The illumination energy efficiency function is oriented to the lighting load. The operating state of electrical appliances is related to the illumination environment of the home area. The illumination energy efficiency function is determined by the brightness of the home area, as shown in Equation (3).

$$E_i^l(L_t) = \begin{cases} 1 & L_t = L_1 \\ 0 & L_t = L_2 \end{cases} \quad (3)$$

Where $E_i^l(L_t)$ represents the illumination energy efficiency function of electrical appliance i ; L_1 represents bright; L_2 represents dark; L_t represents the actual brightness of the user's home area.

3.3. Household energy efficiency assessment

Based on the energy efficiency function of household appliances, use the refined electricity consumption data of each consumer obtained by NILM, including the real-time operating status of each consumer, to conduct household

energy efficiency assessment. Divide household appliances into high-energy-efficiency and low-energy-efficiency appliances. According to the power loss, find high power consumption appliances and provide using opinions.

High energy efficiency means that users have obtained high value and maximized satisfaction while consuming electric energy. Low energy efficiency means that users obtain a certain value or satisfaction in the process of consuming electrical energy, but there is a large electrical energy loss in electrical appliances. Based on the time, temperature and light energy efficiency function, the energy efficiency of different electrical appliances is evaluated, and the electrical appliances are divided into high energy efficiency and low energy efficiency. $U_i(t)$ represents the energy efficiency of electrical appliance i at time t , and high energy efficiency and low energy efficiency are represented by T_h and T_l respectively. The operating state $S_i(t)$ of the electrical appliance is divided into working state and shutdown state, which are represented by T_{on} and T_{off} respectively.

Table 2 shows the energy efficiency evaluation of electrical appliances based on the time energy efficiency function. If the appliance is in working condition and the user is using the appliance, the appliance is in high energy efficiency. When the appliance is off, the appliance is in high energy efficiency. When the electrical appliance is in working condition but the user does not actually need to use the electrical appliance, the electrical appliance is in low energy efficiency.

Table 2. Energy efficiency assessment based on time energy efficiency function

Energy efficiency assessment	Basis of Evaluation	Household appliances
High energy efficiency $U_i(t) = T_h$	$S_i(t) = T_{on}$ $E'_i(t) = 1$	Electrical appliances such as televisions and electric water heaters are in operation and are used by users.
	$S_i(t) = T_{off}$	Electrical appliances such as televisions and electric water heaters are turned off.
Low energy efficiency $U_i(t) = T_l$	$S_i(t) = T_{on}$ $E'_i(t) = 0$	Electrical appliances such as televisions and electric water heaters are in operation and are not actually used by users.

The energy efficiency evaluation of electrical appliances based on the temperature energy efficiency function is shown in Table 3. If the temperature value in the user's home space is outside the user's desired temperature value, and the electrical appliance is in working condition and the actual temperature is guaranteed to be within the temperature value required by the user, the electrical appliance is in high energy efficiency. If the electrical appliance is in a working state and the temperature value in the user's home space is already at the user's desired temperature value, the electrical appliance is in low energy efficiency.

Table 3. Energy efficiency evaluation based on temperature energy efficiency function

Energy efficiency assessment	Basis of Evaluation	Household appliances
High energy efficiency $U_i(t) = T_h$	$S_i(t) = T_{on}$ $E_i^c(t) = 0$	Air conditioners and other electrical appliances are in operation to ensure that the indoor temperature is within the expected value.
	$S_i(t) = T_{off}$	Air conditioners and other electrical appliances are turned off.
Low energy efficiency $U_i(t) = T_l$	$S_i(t) = T_{on}$ $E_i^c(t) = 1$	Air conditioners and other electrical appliances are in operation, and the actual temperature of the home space is within the expected value

The energy efficiency evaluation of electrical appliances based on the light energy efficiency function is shown in Table 4. If the user's home light level is dark, the electrical appliance is in working condition, the electrical appliance is in high energy efficiency. If the user's home light level is bright and the electrical appliances are in working condition, the electrical appliances are in low energy efficiency.

Table 4. Energy efficiency evaluation based on illumination energy efficiency function

Energy efficiency assessment	Basis of Evaluation	Household appliances
High energy efficiency $U_i(t) = T_h$	$S_i(t) = T_{on}$ $E_i^L(t) = 0$	The lighting equipment is in operation to ensure that the room is bright.
	$S_i(t) = T_{off}$	The lighting equipment is off.
Low energy efficiency $U_i(t) = T_l$	$S_i(t) = T_{on}$ $E_i^L(t) = 0$	The lighting equipment is in operation, and the home space is bright.

4. Case analysis

Take the total amount of electrical appliances used by a household on a day and the statistics of the household electricity demand data as an example to evaluate the energy efficiency of the family. The main electrical appliances in this household include air conditioners, electric water heaters, etc. Table 5 shows the fixed use period of the user's electrical appliances, and Table 6 shows the environmental values of the user's electrical appliances. Select the power curve of household total port on a day in summer, and the target period is 17:00-24:00 as shown in Figure 2.

Table 5. The working time of a typical household electrical appliance

Home Space	Household appliances	Active power	Usage time
Dinning room	Drinking fountain	1500W	7:00-8:00; 17:20-22:30
Bedroom or living room	TV	120W	07:00-07:30; 13:00-13:30; 18:00-21:30
Bathroom	Water heater	1700W	21:00-22:00; 6:30-7:30

Table 6. Typical use environment for a user

Household appliances	Active power	Default value	Operating environment
Lighting device	50W~100W	None	Dark
Air conditioners	1200W	18~25°C	>28°C

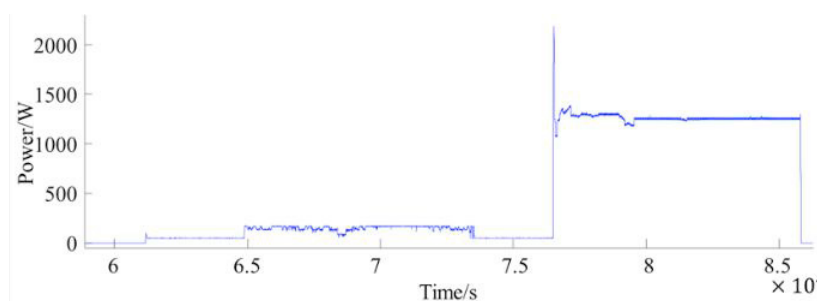


Fig. 2. Total household power curve.

Use NILM technology to obtain detailed electricity consumption data for each electricity consumption. In the target period, the power curve of the TV and air conditioner is shown in Figure 3-(a) and Figure 3-(b). The actual operating period of each appliance can be analyzed, as shown in Table 7.

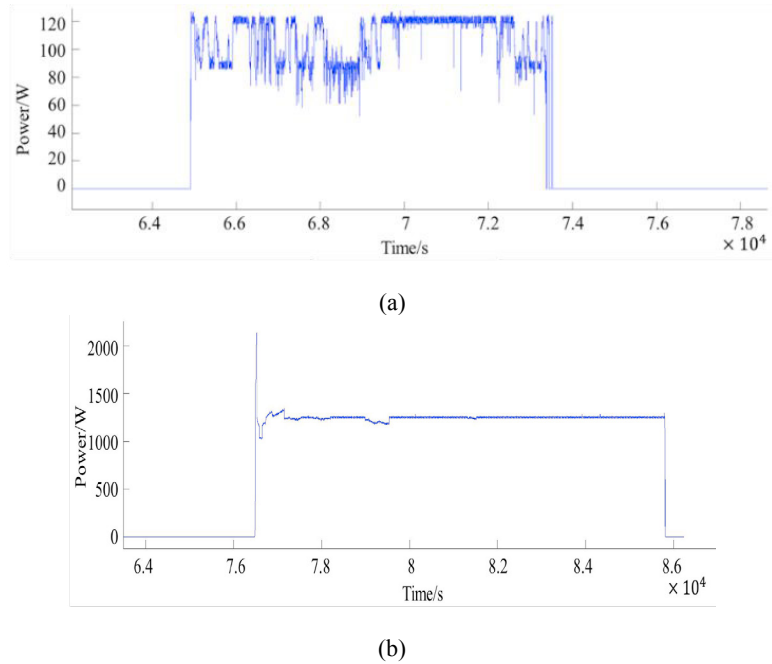


Fig. 3. Active power curve of TV and air conditioning in target period

Table 7. Running time with each appliance

Household appliances	Active power	Actual running time
Air conditioners	1200W	21:17-23:52
TV	120W	18:05-19:55
Lighting device	42W	17:12-22:05

According to the weather data of the day, the lighting environment and temperature environment of the target period of 17:00-24:00 in the user's home space are determined as shown in Figure 6. The blue line is the temperature curve of the user's home space, the red line is the user's desired temperature threshold, which indicates the temperature range is 18-25 degrees Celsius, and the green line indicates the brightness of the user's home.

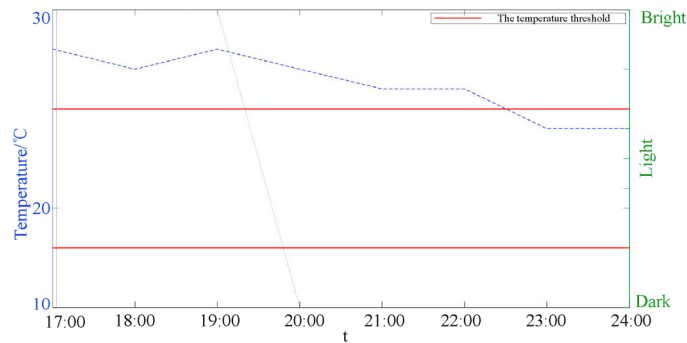


Fig. 4. Light and temperature curve

1) TV energy efficiency evaluation based on time energy efficiency function

In the target period, according to Table 5, the actual use time of the user's TV is 18:00-21:30, and the actual operating time of the TV is 18:05-19:55, so $T_t \in [T_{\min}, T_{\max}]$, $E'_{TV}(t) = 1$, $S_{TV}(t) = T_{on}$, the TV is highly efficient.

2) Air conditioning energy efficiency evaluation based on temperature energy efficiency function

In the target time period, the temperature information of the user's home space is shown in Figure 4. The actual temperature in the home space is higher than the user's expectation from 17:00 to 22:30, so $E^c_{AC}(t) = 0$. The air conditioner is in shutdown state, $S_{AC}(t) = T_{off}$, from 17:00 to 21:17, so the air conditioner is in efficient use. The air conditioner is in operation, $S_{AC}(t) = T_{on}$, from 21:17 to 22:30, so the air conditioner is in efficient use. During 22:30-24:00, the actual temperature in the home space is within the user's expected value, $E^c_{AC}(t) = 1$, but the air conditioner is in operation, $S_{AC}(t) = T_{on}$, from 22:30 to 23:52, so the air conditioner is inefficient during this time.

3) Lighting energy efficiency evaluation based on light energy efficiency function

In the target time period, the illumination information of the user's home space is shown in Figure 4. The illumination in the home space is bright from 17:00 to 19:00, so $E^L_{lighting} = 1$. The lighting equipment is in operation, $S_{lighting}(t) = T_{on}$, from 17:12 to 19:00, so it is inefficient during this time period. From 19:00 to 24:00, the light in the home space is dark, so $E^L_{lighting} = 0$, and the lighting equipment is in operation, $S_{lighting} = T_{on}$, from 19:00 to 22:05, so it is in efficient use during this time period.

In summary, the user's air conditioner is in low energy efficiency at 22:30-23:52, and the lighting equipment is in low energy efficiency from 17:12 to 19:00, so the user should pay attention to the use of air-conditioning and lighting equipment, open windows for ventilation at night, reduce the use of lights during the day, and avoid wasted power.

5. Conclusion

This paper proposes a home energy efficiency assessment method based on NILM. The method fully considers the leading role of users in power activities, and can give the results of energy efficiency evaluation of specific electrical appliances. The cases show that the energy efficiency assessment method can directly lock the users' low energy efficiency appliances. Furthermore, the bad power consumption habits of users can be counted out, and more targeted energy use Suggestions can be put forward to users, to realize energy saving based on ensuring user satisfaction.

References

1. Yu WH, Wu ZG, Zhong H, Bao XY. Analysis of consumer electricity consumption behavior based on electricity big data[J]. *Electromechanical Information*, 2019(20): 92-93.
2. Geng JC, Wang M, Li ZX. Behavior analysis of electric power customers based on big data[J]. *Henan Electric Power Technology*, 2017(03): 33-35.
3. Xu LJ, Zhang XD, Wang LJ. User classification and electricity consumption behavior analysis based on cluster analysis[J]. *Shanxi Electric Power*, 2016(04): 23-27.
4. Qu CY, Wang C, Pan F. Household Electric Energy Efficiency Evaluation Method in Smart Electricity Environment[J]. *Advanced Technology of Electrical Engineering and Energy*, 2015, 34(10): 61-67.
5. Lu J, Zhu YP, Peng WH, Sun Y. Strategy for optimizing characteristics of intelligent electricity user behavior analysis[J]. *Automation of Electric Power Systems*, 2017, 41(05): 58-63+83.
6. Shi TG, Wang JB, Kou LF, et al. Smart home appliance management and control scheme based on dynamic home appliance priority[J]. *Electrical Measurement & Instrumentation*, 2017, 55(23).
7. Wang F, Li KM. Research and Application of User Energy Efficiency Analysis System[J]. *Electric Power Demand Side Management*, 2017(04): 43-46.
8. Zhao XL, He GY. The concept of electricity utility of household appliances and its evaluation method[J]. *Automation of Electric Power Systems*, 2016, v.40; No.575(01): 53-59.
9. Liu B, Luan WP, Yu YX. Dynamic time warping based non-intrusive load transient identification[J]. *Applied Energy*, 2017; 195: 634-645.
10. Liu B, Yu YX, Luan WP, et al. An unsupervised electrical appliance modeling framework for non-intrusive load monitoring[C]. *2017 IEEE Power & Energy Society General Meeting*, 2017: 1-5.
11. Chen ZH, Ma Li E, Liu P, Li Z. Electric vehicle development in China: A charging behavior and power sector supply balance analysis[J]. *Chemical Engineering Research and Design*, 2018, 131.